

Statistics Primer

Part of **Data Science Starter** · Estimated time: 40 min

Introduction

You don't need advanced math to ask good questions of data — a small set of statistical ideas covers most everyday cases.

Core ideas

- The mean (average) is sensitive to outliers; the median (the middle value) is not. When a dataset has extreme values, the median often tells a truer story.
- Spread — variance and standard deviation — tells you how reliable an average actually is. Two datasets can have the same mean and look completely different once you account for spread.
- Correlation measures how strongly two variables move together, from -1 to 1, but a strong correlation never proves that one variable causes the other.
- A classic example: ice cream sales and drowning incidents both rise in summer — they're correlated, but neither causes the other; both are driven by warm weather.

Example

```
import pandas as pd
print(df["revenue"].mean(), df["revenue"].median())
print(df[["revenue", "marketing_spend"]].corr())
```

Try it yourself

Take any two numeric columns in a dataset you have access to and compute their correlation. Write one sentence about what a confounding factor might be if the correlation turns out to be strong.

Further reading

- [Khan Academy — Statistics & Probability](https://www.khanacademy.org/math/statistics-probability)

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- [Online Statistics Education \(OpenStax/Rice\)](https://online.statbook.com/)

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ModuloTrainer reading material — return to the app to mark this module complete.